Web scraping and Parallel Processing

Greg Ridgeway Ruth Moyer

2026-06-22

Table of contents

1 Introduction

At the end of our discussion about regular expressions, we introduced the concept of web scraping. Not all online data is in a tidy, downloadable format such as a .csv or .RData file. Yet, patterns in the underlying HTML code and regular expressions together provide a valuable way to "scrape" data off of a webpage. Here, we are going to work through an example of web scraping. We are going to get data on ticket sales of every movie, for every day going back to 2010.

As a preliminary matter, some R packages, such as `rvest` and `chromote`, can help with web scraping. Eventually you may wish to explore those packages. For now, we are going to work with basic fundamentals so that you have the most flexibility to extract data from most websites.

First, you will need to make sure that you can access the underlying HTML code for the webpage that you want to scrape. In most browsers you can simply right-click on a webpage and then click “View Page Source.” If you are using Microsoft Edge, you can right-click on the webpage, click “View Source,” and then look at the “Debugger” tab. In Safari select “Settings,” choose the “Advanced” tab, check “Show Develop menu,” and then whenever viewing a page you can right-click and select “Show Page Source.”

Have a look at the webpage https://www.the-numbers.com/box-office-chart/daily/2025/07/04. This page contains information about the movies that were shown in theaters on July 4, 2025 and the amount of money (in dollars) that each of those movies grossed that day.

Have a look at the HTML code by looking at the page source for this page using the methods described above. The first 10 lines should look something like this:

```
<!DOCTYPE html>
<html xmlns:og="https://ogp.me/ns#">
<head>
<link rel='icon' href='https://www.the-numbers.com/site-images/favicon.ico'>
<script async src="https://www.googletagmanager.com/gtag/js?id=G-5K2DT3XQN5"></script>
<script>window.dataLayer = window.dataLayer || []; function gtag() { dataLayer.push(argument
5K2DT3XQN5');</script>
<meta http-equiv="PICS-Label" content='(PICS-1.1 "https://www.icra.org/ratingsv02.html" l ge
numbers.com/" r (cb 1 lz 1 nz 1 oz 1 vz 1) "https://www.rsac.org/ratingsv01.html" l gen true
numbers.com/" r (n 0 s 0 v 0 l 0))'>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8">
<meta name="format-detection" content="telephone=no">   <!-- for apple mobile -->
<meta name=viewport content="width=device-width, initial-scale=1">
```

This is all HTML code to set up the page. If you scroll down a few hundred lines, you will find code that looks like this:

```
<tbody>
<tr>
<td class="data">1</td>
<td class="data">(1)</td>
<td><b><a href="/movie/Jurassic-World-Rebirth-(2025)">Jurassic World Rebirth</a></b></td>
<td class="data">$26,235,450</td>
<td class="data chart_up d">+4%</td>
<td class="data Daily"> </td>
<td class="data">4,308</td>
<td class="data">$6,090</td>
<td class="data">$82,040,080</td>
<td class="data">3</td>
</tr>
```

I see *Jurassic World Rebirth* and *F1: The Movie*. In addition to the movie name, there are ticket sales, number of theaters, and more. It is all wrapped in a lot of HTML code to make it look pretty on a web page, but for our purposes we just want to pull those numbers out.

`scan()` is a basic R function for reading in text, from the keyboard, from files, from the web, … however data might arrive. Giving `scan()` a URL causes `scan()` to pull down the HTML code for that page and return it to you. Let's try one page of movie data. `what=""` tells `scan()` to expect plain text and `sep="\n"` tells `scan()` to separate each element when it reaches a line feed character, signaling the end of a line.

Here I have wrapped `scan()` inside `try()`. More on this later. Briefly, this `try()` inside the `repeat{}` loop will allow for errors, like the website being unavailable for a moment. If `scan()` cannot reach the website, then `Sys.sleep(10)` will pause for 10 seconds before trying again.

```
library(dplyr)
repeat
{
   a <- try(scan("https://www.the-numbers.com/box-office-chart/daily/2025/07/04",
             what="",sep="\n"))
   if(!inherits(a,"try-error")) break
   Sys.sleep(10)
}
# examine the first few lines
a[1:5]
```

```
[1] "<!DOCTYPE html>"
[2] "<html xmlns:og=\"https://ogp.me/ns#\">"
[3] "<head>"
[4] "<link rel='icon' href='https://www.the-numbers.com/site-images/favicon.ico'>"
[5] "<script async src=\"https://www.googletagmanager.com/gtag/js?id=G-
5K2DT3XQN5\"></script>"
```

Some websites are more complex or use different text encoding. On those websites `scan()` produces unintelligible text. The `GET()` function from the `httr` package can sometimes resolve this.

```
library(httr)
resp <- GET("https://www.the-numbers.com/box-office-chart/daily/2025/07/04")
a1 <- content(resp, as="text")
a1 <- strsplit(a1,"\n")[[1]]
```

Also, some Mac users will encounter snags with both of these methods and receive "403 Forbidden" errors while their Mac colleague right next to them on the same network will not.

I have not figured out why this happens, but have found that making R masquerade as a different browser sometimes works.

```
resp <- GET("https://www.the-numbers.com/box-office-chart/daily/2025/07/04",
            user_agent("Mozilla/5.0 (Macintosh; Intel Mac OS X 10_6_8) AppleWebKit/537.13+ (
a1 <- content(resp, as="text")
a1 <- strsplit(a1,"\n")[[1]]
```

2 Scraping one page

Now that we have stored in the variable `a` the HTML code for one day's movie data in R, let's apply some regular expressions to extract the data. The HTML code includes a lot of lines that do not involve data that interests us. There is code for making the page look nice and code for presenting advertisements. Let's start by finding the lines that have the movie names in them.

Going back to the HTML code, I noticed that the lines with *Jurassic World Rebirth* and *F1: The Movie* both have the sequence of characters `href="/movie/`. By finding a pattern of characters that always appears on movie title lines, we can use it to grep the lines we want. Let's find every line that has `href="/movie/` in it.

```
i <- grep('href="/movie/', a)
i
```

```
 [1] 198 210 222 234 246 258 270 282 294 306 318 330 342 354 366 378 390 402 414
[20] 426 438 450 462 474 486 498 510 522 534 546 574 580 586 592 598 604 610 616
[39] 622 628 634 640 646 652 658 664 670 676 682 688 694 700 706 712 718 724 730
[58] 736 742 748
```

These are the line numbers that, if the pattern holds, contain our movie titles. Note that the numbers you get on your computer might be a little different from the line numbers shown here. Even if you run this code on a different day, you might get different line numbers because some of the code, code for advertisements in particular, can frequently change.

Let's see what these lines of HTML code look like.

```
a[i]
```

```
 [1] "<td><b><a href=\"/movie/Jurassic-World-Rebirth-(2025)\">Jurassic World Rebirth</a></b>
 [2] "<td><b><a href=\"/movie/F1-The-Movie-(2025)\">F1: The Movie</a></b></td>"
 [3] "<td><b><a href=\"/movie/How-to-Train-Your-Dragon-(2025)\">How to Train Your Dragon</a>
```

```
 [4] "<td><b><a href=\"/movie/Elio-(2025)\">Elio</a></b></td>"
 [5] "<td><b><a href=\"/movie/28-Years-Later-(2025)\">28 Years Later</a></b></td>"
 [6] "<td><b><a href=\"/movie/M3GAN-2-0-(2025)\">M3GAN 2.0</a></b></td>"
 [7] "<td><b><a href=\"/movie/Lilo-and-Stitch-(2025)\">Lilo &amp; Stitch</a></b></td>"
 [8] "<td><b><a href=\"/movie/Mission-Impossible-The-Final-Reckoning-(2025)\">Mission: Imposs
 [9] "<td><b><a href=\"/movie/This-is-Spinal-Tap-(1984)\">This is Spinal Tap</a></b></td>"
[10] "<td><b><a href=\"/movie/Materialists-(2025)\">Materialists</a></b></td>"
[11] "<td><b><a href=\"/movie/Sardaar-Ji-3-(2025-India)\">Sardaar Ji 3</a></b></td>"
[12] "<td><b><a href=\"/movie/From-the-World-of-John-Wick-Ballerina-(2025)\">From the World o
[13] "<td><b><a href=\"/movie/Phoenician-Scheme-The-(2025)\">The Phoenician Scheme</a></b></t
[14] "<td><b><a href=\"/movie/Karate-Kid-Legends-(2025)\">Karate Kid: Legends</a></b></td>"
[15] "<td><b><a href=\"/movie/Final-Destination-Bloodlines-(2025)\">Final Destination: Bloodl
[16] "<td><b><a href=\"/movie/Life-of-Chuck-The-(2025)\">The Life of Chuck</a></b></td>"
[17] "<td><b><a href=\"/movie/Sinners-(2025)\">Sinners</a></b></td>"
[18] "<td><b><a href=\"/movie/Sorry-Baby-(2025)\">Sorry, Baby</a></b></td>"
[19] "<td><b><a href=\"/movie/Thunderbolts-(2025)\">Thunderbolts*</a></b></td>"
[20] "<td><b><a href=\"/movie/Last-Rodeo-The-(2025)\">The Last Rodeo</a></b></td>"
[21] "<td><b><a href=\"/movie/Friendship-(2025)\">Friendship</a></b></td>"
[22] "<td><b><a href=\"/movie/Bring-Her-Back-(2025)\">Bring Her Back</a></b></td>"
[23] "<td><b><a href=\"/movie/Jane-Austen-a-gache-ma-vie-(2025-France)\">Jane Austen Wrecked
[24] "<td><b><a href=\"/movie/Hearts-of-Darkness-A-Filmmakers-Apocalypse\">Hearts of Darkness
[25] "<td><b><a href=\"/movie/Hot-Milk-(2025-United-Kingdom)\">Hot Milk</a></b></td>"
[26] "<td><b><a href=\"/movie/Unholy-Trinity-The-(2025)\">The Unholy Trinity</a></b></td>"
[27] "<td><b><a href=\"/movie/Ran-(1985-Japan)\">Ran</a></b></td>"
[28] "<td><b><a href=\"/movie/Dragon-Heart-Adventures-Beyond-This-World-
(2025-Japan)\">Dragon Heart - Adventures Beyond This World</a></b></td>"
[29] "<td><b><a href=\"/movie/Dangerous-Animals-(2025-Australia)\">Dangerous Animals</a></b>
[30] "<td><b><a href=\"/movie/King-of-Kings-The-(2025-South-Korea)\">The King of Kings</a></
[31] "<td><b><a href=\"/movie/Jurassic-World-Rebirth-(2025)\">Jurassic World Rebirth</a></b>
[32] "<td><b><a href=\"/movie/F1-The-Movie-(2025)\">F1: The Movie</a></b></td>"
[33] "<td><b><a href=\"/movie/How-to-Train-Your-Dragon-(2025)\">How to Train Your Dragon</a>
[34] "<td><b><a href=\"/movie/Elio-(2025)\">Elio</a></b></td>"
[35] "<td><b><a href=\"/movie/28-Years-Later-(2025)\">28 Years Later</a></b></td>"
[36] "<td><b><a href=\"/movie/M3GAN-2-0-(2025)\">M3GAN 2.0</a></b></td>"
[37] "<td><b><a href=\"/movie/Lilo-and-Stitch-(2025)\">Lilo &amp; Stitch</a></b></td>"
[38] "<td><b><a href=\"/movie/Mission-Impossible-The-Final-Reckoning-(2025)\">Mission: Imposs
[39] "<td><b><a href=\"/movie/This-is-Spinal-Tap-(1984)\">This is Spinal Tap</a></b></td>"
[40] "<td><b><a href=\"/movie/Materialists-(2025)\">Materialists</a></b></td>"
[41] "<td><b><a href=\"/movie/Sardaar-Ji-3-(2025-India)\">Sardaar Ji 3</a></b></td>"
[42] "<td><b><a href=\"/movie/From-the-World-of-John-Wick-Ballerina-(2025)\">From the World o
[43] "<td><b><a href=\"/movie/Phoenician-Scheme-The-(2025)\">The Phoenician Scheme</a></b></t
[44] "<td><b><a href=\"/movie/Karate-Kid-Legends-(2025)\">Karate Kid: Legends</a></b></td>"
[45] "<td><b><a href=\"/movie/Final-Destination-Bloodlines-(2025)\">Final Destination: Bloodl
```

```
[46] "<td><b><a href=\"/movie/Life-of-Chuck-The-(2025)\">The Life of Chuck</a></b></td>"
[47] "<td><b><a href=\"/movie/Sinners-(2025)\">Sinners</a></b></td>"
[48] "<td><b><a href=\"/movie/Sorry-Baby-(2025)\">Sorry, Baby</a></b></td>"
[49] "<td><b><a href=\"/movie/Thunderbolts-(2025)\">Thunderbolts*</a></b></td>"
[50] "<td><b><a href=\"/movie/Last-Rodeo-The-(2025)\">The Last Rodeo</a></b></td>"
[51] "<td><b><a href=\"/movie/Friendship-(2025)\">Friendship</a></b></td>"
[52] "<td><b><a href=\"/movie/Bring-Her-Back-(2025)\">Bring Her Back</a></b></td>"
[53] "<td><b><a href=\"/movie/Jane-Austen-a-gache-ma-vie-(2025-France)\">Jane Austen Wrecked
[54] "<td><b><a href=\"/movie/Hearts-of-Darkness-A-Filmmakers-Apocalypse\">Hearts of Darknes
[55] "<td><b><a href=\"/movie/Hot-Milk-(2025-United-Kingdom)\">Hot Milk</a></b></td>"
[56] "<td><b><a href=\"/movie/Unholy-Trinity-The-(2025)\">The Unholy Trinity</a></b></td>"
[57] "<td><b><a href=\"/movie/Ran-(1985-Japan)\">Ran</a></b></td>"
[58] "<td><b><a href=\"/movie/Dragon-Heart-Adventures-Beyond-This-World-
(2025-Japan)\">Dragon Heart - Adventures Beyond This World</a></b></td>"
[59] "<td><b><a href=\"/movie/Dangerous-Animals-(2025-Australia)\">Dangerous Animals</a></b>
[60] "<td><b><a href=\"/movie/King-of-Kings-The-(2025-South-Korea)\">The King of Kings</a></
```

Double-checking and indeed the first line here is *Jurassic World Rebirth* and the last line is *The King of Kings*. This matches what is on the web page. We now are quite close to having a list of movies that played in theaters on July 4, 2025. However, as you can see, we have a lot of excess symbols and HTML code to eliminate before we can have a neat list of movie names.

HTML tags always have the form `<some code here>`. Therefore, we should remove any text between a less than and greater than symbol. Here is a regular expression that looks for a `<` followed by any number of characters that are not `>`, and then a closing `>`… and `gsub()` will delete them.

```
gsub("<[^>]*>", "", a[i])
```

```
 [1] "Jurassic World Rebirth"
 [2] "F1: The Movie"
 [3] "How to Train Your Dragon"
 [4] "Elio"
 [5] "28 Years Later"
 [6] "M3GAN 2.0"
 [7] "Lilo &amp; Stitch"
 [8] "Mission: Impossible-The Final Reckoning"
 [9] "This is Spinal Tap"
[10] "Materialists"
[11] "Sardaar Ji 3"
[12] "From the World of John Wick: Ballerina"
```

[13] "The Phoenician Scheme"
[14] "Karate Kid: Legends"
[15] "Final Destination: Bloodlines"
[16] "The Life of Chuck"
[17] "Sinners"
[18] "Sorry, Baby"
[19] "Thunderbolts*"
[20] "The Last Rodeo"
[21] "Friendship"
[22] "Bring Her Back"
[23] "Jane Austen Wrecked My Life"
[24] "Hearts of Darkness: A Filmmaker's Apocalypse"
[25] "Hot Milk"
[26] "The Unholy Trinity"
[27] "Ran"
[28] "Dragon Heart - Adventures Beyond This World"
[29] "Dangerous Animals"
[30] "The King of Kings"
[31] "Jurassic World Rebirth"
[32] "F1: The Movie"
[33] "How to Train Your Dragon"
[34] "Elio"
[35] "28 Years Later"
[36] "M3GAN 2.0"
[37] "Lilo & Stitch"
[38] "Mission: Impossible-The Final Reckoning"
[39] "This is Spinal Tap"
[40] "Materialists"
[41] "Sardaar Ji 3"
[42] "From the World of John Wick: Ballerina"
[43] "The Phoenician Scheme"
[44] "Karate Kid: Legends"
[45] "Final Destination: Bloodlines"
[46] "The Life of Chuck"
[47] "Sinners"
[48] "Sorry, Baby"
[49] "Thunderbolts*"
[50] "The Last Rodeo"
[51] "Friendship"
[52] "Bring Her Back"
[53] "Jane Austen Wrecked My Life"
[54] "Hearts of Darkness: A Filmmaker's Apocalypse"
[55] "Hot Milk"

```
[56] "The Unholy Trinity"
[57] "Ran"
[58] "Dragon Heart - Adventures Beyond This World"
[59] "Dangerous Animals"
[60] "The King of Kings"
```

Perfect! Now we just have movie names. You will see some movie names have strange symbols, like `…` or `'`. These are the HTML codes for horizontal ellipses "…" and a smart apostrophe. These make the text look prettier on a webpage, but you might need to do more work with `gsub()` if it is important that these movie names look right.

Let's put these movie names in a data frame, `data0`. This data frame currently has only one column.

```r
data0 <- data.frame(movie=gsub("<[^>]*>", "", a[i]))
```

Now we also want to get the daily gross for each movie. Let's take another look at the HTML code for *Jurassic World Rebirth*.

```r
a[i[1] + 0:8]
```

```
[1] "<td><b><a href=\"/movie/Jurassic-World-Rebirth-(2025)\">Jurassic World Rebirth</a></b><
[2] "<td class=\"data\">$26,235,450</td>"
[3] "<td class=\"data chart_up d\">+4%</td>"
[4] "<td class=\"data Daily\"> </td>"
[5] "<td class=\"data\">4,308</td>"
[6] "<td class=\"data\">$6,090</td>"
[7] "<td class=\"data\">$82,040,080</td>"
[8] "<td class=\"data\">3</td>"
[9] "</tr>"
```

Note that the movie gross is one line after the movie name. It turns out that this is consistent for all movies. Since `i` has the line numbers for the movie names, then `i+1` must be the line numbers containing the daily gross.

```r
a[i+1]
```

```
 [1] "<td class=\"data\">$26,235,450</td>"
 [2] "<td class=\"data\">$6,960,390</td>"
 [3] "<td class=\"data\">$2,909,540</td>"
 [4] "<td class=\"data\">$1,500,046</td>"
```

```
 [5] "<td class=\"data\">$1,106,889</td>"
 [6] "<td class=\"data\">$972,945</td>"
 [7] "<td class=\"data\">$962,789</td>"
 [8] "<td class=\"data\">$851,931</td>"
 [9] "<td class=\"data\">$431,360</td>"
[10] "<td class=\"data\">$354,396</td>"
[11] "<td class=\"data chart_estimate\">$220,000</td>"
[12] "<td class=\"data\">$201,834</td>"
[13] "<td class=\"data\">$116,865</td>"
[14] "<td class=\"data\">$100,872</td>"
[15] "<td class=\"data\">$85,239</td>"
[16] "<td class=\"data\">$73,304</td>"
[17] "<td class=\"data\">$41,283</td>"
[18] "<td class=\"data\">$32,621</td>"
[19] "<td class=\"data\">$14,381</td>"
[20] "<td class=\"data\">$12,136</td>"
[21] "<td class=\"data\">$4,165</td>"
[22] "<td class=\"data\">$4,158</td>"
[23] "<td class=\"data\">$2,798</td>"
[24] "<td class=\"data\">$2,274</td>"
[25] "<td class=\"data\">$796</td>"
[26] "<td class=\"data\">$680</td>"
[27] "<td class=\"data\">$629</td>"
[28] "<td class=\"data\">$606</td>"
[29] "<td class=\"data\">$506</td>"
[30] "<td class=\"data\">$143</td>"
[31] "<td class=\"data\">$26,235,450</td>"
[32] "<td class=\"data\">$6,960,390</td>"
[33] "<td class=\"data\">$2,909,540</td>"
[34] "<td class=\"data\">$1,500,046</td>"
[35] "<td class=\"data\">$1,106,889</td>"
[36] "<td class=\"data\">$972,945</td>"
[37] "<td class=\"data\">$962,789</td>"
[38] "<td class=\"data\">$851,931</td>"
[39] "<td class=\"data\">$431,360</td>"
[40] "<td class=\"data\">$354,396</td>"
[41] "<td class=\"data chart_estimate\">$220,000</td>"
[42] "<td class=\"data\">$201,834</td>"
[43] "<td class=\"data\">$116,865</td>"
[44] "<td class=\"data\">$100,872</td>"
[45] "<td class=\"data\">$85,239</td>"
[46] "<td class=\"data\">$73,304</td>"
[47] "<td class=\"data\">$41,283</td>"
```

```
[48] "<td class=\"data\">$32,621</td>"
[49] "<td class=\"data\">$14,381</td>"
[50] "<td class=\"data\">$12,136</td>"
[51] "<td class=\"data\">$4,165</td>"
[52] "<td class=\"data\">$4,158</td>"
[53] "<td class=\"data\">$2,798</td>"
[54] "<td class=\"data\">$2,274</td>"
[55] "<td class=\"data\">$796</td>"
[56] "<td class=\"data\">$680</td>"
[57] "<td class=\"data\">$629</td>"
[58] "<td class=\"data\">$606</td>"
[59] "<td class=\"data\">$506</td>"
[60] "<td class=\"data\">$143</td>"
```

Again we need to strip out the HTML tags. We will also remove the dollar signs and commas so that R will recognize it as a number. We will add this to `data0` also.

```
data0$gross <- as.numeric(gsub("<[^>]*>|[$,]", "", a[i+1]))
```

Take a look at the webpage and compare it to the dataset you have now created. All the values should now match.

```
head(data0)
```

```
                    movie    gross
1   Jurassic World Rebirth 26235450
2            F1: The Movie  6960390
3 How to Train Your Dragon  2909540
4                     Elio  1500046
5           28 Years Later  1106889
6               M3GAN 2.0   972945
```

```
tail(data0)
```

```
                                         movie gross
55                                    Hot Milk   796
56                          The Unholy Trinity   680
57                                         Ran   629
58 Dragon Heart - Adventures Beyond This World   606
59                           Dangerous Animals   506
60                            The King of Kings   143
```

2.1 Movies with no titles

Some movies have missing titles. Have a look at the February 21, 2011 movies.

```
repeat
{
   a <- try(scan("https://www.the-numbers.com/box-office-chart/daily/2011/02/21",
             what="",sep="\n"))
   if(!inherits(a,"try-error")) break
   Sys.sleep(10)
}
i <- grep("Genesis-Code", a)[1]
a[-10:10 + i]
```

```
 [1] "<td class=\"data  d\"> </td>"
 [2] "<td class=\"data Daily\"> </td>"
 [3] "<td class=\"data\">12</td>"
 [4] "<td class=\"data\">$155</td>"
 [5] "<td class=\"data\">$951,246</td>"
 [6] "<td class=\"data\">67</td>"
 [7] "</tr>"
 [8] "<tr>"
 [9] "<td class=\"data\">62</td>"
[10] "<td class=\"data\">(-)</td>"
[11] "<td><b><a href=\"/movie/Genesis-Code-The-(2010)\"></a></b></td>"
[12] "<td class=\"data chart_estimate\">$1,800</td>"
[13] "<td class=\"data  d\"> </td>"
[14] "<td class=\"data Daily\"> </td>"
[15] "<td class=\"data\">17</td>"
[16] "<td class=\"data\">$106</td>"
[17] "<td class=\"data\">$20,300</td>"
[18] "<td class=\"data\">4</td>"
[19] "</tr>"
[20] "<tr>"
[21] "<td class=\"data\">63</td>"
```

Note that the line for *The Genesis Code* has the movie title in the `href` attribute, but no movie title is between the `<a></a>` tags. In these cases let's pull the movie title from the `href` attribute.

```r
i <- grep('href=\"/movie/',a)
data0 <- data.frame(movie = gsub("<[^>]*>","",a[i]),
                    gross = as.numeric(gsub("<[^>]*>|[,$]","",a[i+1])))
# which ones are blank?
j <- which(data0$movie=="")
a[i[j]]
```

```
[1] "<td><b><a href=\"/movie/Genesis-Code-The-(2010)\"></a></b></td>"
[2] "<td><b><a href=\"/movie/Rauber-Der\"></a></b></td>"
[3] "<td><b><a href=\"/movie/Genesis-Code-The-(2010)\"></a></b></td>"
[4] "<td><b><a href=\"/movie/Rauber-Der\"></a></b></td>"
```

```r
# test regex to pull movie title
gsub('.*/movie/([^"]*)".*', "\\1", a[i[j]])
```

```
[1] "Genesis-Code-The-(2010)" "Rauber-Der"
[3] "Genesis-Code-The-(2010)" "Rauber-Der"
```

```r
# replace empty movie names
data0$movie[j] <- gsub('.*/movie/([^"]*)".*', "\\1", a[i[j]]) |>
  gsub("-", " ", x=_)
```

3 Scraping Multiple Pages

We have now successfully scraped data for one day. This is usually the hardest part. But if we have R code that can correctly scrape one day's worth of data *and* the website is consistent across days, then it is simple to adapt our code to work for *all* days. So let's get all movie data from January 1, 2010 through July 31, 2026. That means we are going to be web scraping 6,056 pages of data.

First note that the URL for July 4, 2025 was

`https://www.the-numbers.com/box-office-chart/daily/2025/07/04`

We can extract data from any other date by using the same URL, but changing the ending to match the date that we want. Importantly, the 07 and the 04 in the URL must have leading zeros for the URL to return the correct page.

To start, let's make a list of all the dates that we intend to scrape.

```r
library(lubridate)
# create a sequence of all days to scrape
dates2scrape <- seq(ymd("2010-01-01"), ymd("2026-07-31"), by="days")
```

Now `dates2scrape` contains a collection of all the dates with movie data that we wish to scrape.

```r
dates2scrape[1:5]
```

```
[1] "2010-01-01" "2010-01-02" "2010-01-03" "2010-01-04" "2010-01-05"
```

```r
# gsub() to change - to / matching appearance of thenumbers.com URL
gsub("-", "/", dates2scrape[1:5])
```

```
[1] "2010/01/01" "2010/01/02" "2010/01/03" "2010/01/04" "2010/01/05"
```

Our plan is to construct a for-loop within which we will construct a URL from `dates2scrape`, pull down the HTML code from that URL, scrape the movie data into a data frame, and then combine each day's data frame into one data frame with all of the movie data. First we create a list that will contain each day's data frame.

```r
results <- vector("list", length(dates2scrape))
```

On iteration `i` of our for loop we will store that day's movie data frame in `results[[i]]`. The following for loop can take several minutes to run and its speed will depend on your network connection and how responsive the web site is. Before running the entire for loop, it may be a good idea to temporarily set the dates to a short period of time (e.g., a month or two) just to verify that your code is functioning properly. Once you have concluded that the code is doing what you want it to do, you can set the dates so that the for loop runs for the entire analysis period.

This loop takes about an hour to pull all the data.

```r
timeStart <- Sys.time() # record the starting time
for(iDate in 1:length(dates2scrape))
{
   # useful to know how much is left to go... periodic messages
   if((iDate <= 10) || (iDate %% 30 == 0))
   {
      cat(as.character(dates2scrape[iDate]), "\n", file=stderr())
```

```r
  }

  # construct URL
  urlText <- paste0("https://www.the-numbers.com/box-office-chart/daily/",
                    gsub("-", "/", dates2scrape[iDate]))

  # read in the HTML code
  tries <- 0
  repeat
  {
     tries <- tries + 1
     a <- try(scan(urlText, what="", sep="\n",
                   fileEncoding="UTF-8",
                   quiet = TRUE))
     if(!inherits(a, "try-error") || tries >= 5) break
     Sys.sleep(10)
  }
  # skip this date if all 5 attempts failed
  if(inherits(a, "try-error")) next

  # find movies
  i <- grep('href="/movie/', a)
  # keep only those in "chart-desktop" table... skip "chart-mobile"
  i <- i[i < grep("chart-mobile",a)[1]]

  # get movie names and gross
  data0 <- data.frame(movie = gsub("<[^>]*>", "", a[i]),
                      gross = as.numeric(gsub("<[^>]*>|[$,]","",a[i+1])),
                      date  = dates2scrape[iDate])

  # replace empty movie names
  j <- which(data0$movie=="")
  data0$movie[j] <- gsub('.*/movie/([^"]*)".*', "\\1", a[i[j]]) |>
     gsub("-", " ", x=_)

  results[[iDate]] <- data0
}
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
```

```
'https://www.the-numbers.com/box-office-chart/daily/2026/06/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/22': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/23': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/24': HTTP status
```

```
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/25': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/26': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/27': HTTP status
```

```
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/27': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/28': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/29': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/30': HTTP status
```

```
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/06/30': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/01': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/01': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/01': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/01': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/01': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/02': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/02': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/02': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/02': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/02': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/03': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/03': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/03': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/03': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/03': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/04': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/04': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/04': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/04': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/04': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/05': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/05': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/05': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/05': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/05': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/06': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/06': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/06': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/06': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/06': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/07': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/07': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/07': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/07': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/07': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/08': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/08': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/08': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/08': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/08': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/09': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/09': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/09': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/09': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/09': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/10': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/10': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/10': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
```

```
'https://www.the-numbers.com/box-office-chart/daily/2026/07/10': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/10': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/11': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/11': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/11': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/11': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/11': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/12': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/12': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/12': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/12': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/12': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/13': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
```

```
'https://www.the-numbers.com/box-office-chart/daily/2026/07/13': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/13': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/13': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/13': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/14': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/14': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/14': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/14': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/14': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/15': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/15': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/15': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/15': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/15': HTTP status
```

```
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/16': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/16': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/16': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/16': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/16': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/17': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/17': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/17': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/17': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/17': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/18': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/18': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/18': HTTP status
```

```
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/18': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/18': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/19': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/19': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/19': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/19': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/19': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/20': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/20': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/20': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/20': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/20': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/21': HTTP status
```

```
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/21': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/21': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/21': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/21': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/22': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/22': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/23': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/23': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/23': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/24': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/24': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/25': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/25': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/26': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/26': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/26': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/27': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/27': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/28': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/28': HTTP status
was '404 Not Found'
```

```
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/29': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/29': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/30': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/30': HTTP status
was '404 Not Found'

Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/31': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/31': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/31': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
```

```
'https://www.the-numbers.com/box-office-chart/daily/2026/07/31': HTTP status
was '404 Not Found'
Warning in file(file, "r", encoding = fileEncoding): cannot open URL
'https://www.the-numbers.com/box-office-chart/daily/2026/07/31': HTTP status
was '404 Not Found'
```

```r
# calculate how long it took
timeEnd <- Sys.time()
timeEnd-timeStart
```

Let's look at the first 3 lines of the first and last 3 days.

```r
# first 6 rows of first 3 days
results |> head(3) |> lapply(head)
```

```
[[1]]
                                  movie    gross       date
1                                Avatar 25274008 2010-01-01
2                       Sherlock Holmes 14889882 2010-01-01
3 Alvin and the Chipmunks: The Squeakquel 12998264 2010-01-01
4                      It's Complicated  7127425 2010-01-01
5                        The Blind Side  4554779 2010-01-01
6                         Up in the Air  4112263 2010-01-01

[[2]]
                                  movie    gross       date
1                                Avatar 25835551 2010-01-02
2                       Sherlock Holmes 14373564 2010-01-02
3 Alvin and the Chipmunks: The Squeakquel 14373273 2010-01-02
4                      It's Complicated  7691535 2010-01-02
5                        The Blind Side  4997659 2010-01-02
6                         Up in the Air  4457565 2010-01-02

[[3]]
                                  movie    gross       date
1                                Avatar 17381129 2010-01-03
2 Alvin and the Chipmunks: The Squeakquel  7818116 2010-01-03
3                       Sherlock Holmes  7349035 2010-01-03
4                      It's Complicated  3984005 2010-01-03
5                        The Blind Side  2360311 2010-01-03
6              The Princess and the Frog  2264727 2010-01-03
```

```r
# first 6 rows of last 3 days
results |> tail(3) |> lapply(head)
```

```
[[1]]
NULL

[[2]]
NULL

[[3]]
NULL
```

Looks like we got them all. Now let's combine them into one big data frame. `bind_rows()` takes a list of data frames, like `results[[1]]`, `results[[2]]`, …, and stacks them all on top of each other.

```r
movieData <- bind_rows(results)

# check that the number of rows and dates seem reasonable
nrow(movieData)
```

```
[1] 235874
```

```r
range(movieData$date)
```

```
[1] "2010-01-01" "2026-06-21"
```

```r
head(movieData)
```

```
                                  movie    gross       date
1                                Avatar 25274008 2010-01-01
2                       Sherlock Holmes 14889882 2010-01-01
3 Alvin and the Chipmunks: The Squeakquel 12998264 2010-01-01
4                      It's Complicated  7127425 2010-01-01
5                        The Blind Side  4554779 2010-01-01
6                         Up in the Air  4112263 2010-01-01
```

```r
tail(movieData)
```

```
                               movie  gross       date
235869           Stop! That! Train! 133797 2026-06-21
235870                     Pressure 120000 2026-06-21
235871 The Super Mario Galaxy Movie  35000 2026-06-21
235872               I Love Boosters  24871 2026-06-21
235873                  Unidentified   4000 2026-06-21
235874                    Deep Water    217 2026-06-21
```

If you ran that for-loop to gather 15 years worth of data, most likely you walked away from your computer to do something more interesting than watch its progress. In these situations, I like to send myself a text message when it is complete. The `emayili` package is a convenient way to send yourself an email or text. If you fill it in with your email, username, and gmail app password, the following code will send you an email or text message when the script reaches this point.

```r
library(emayili)

# https://myaccount.google.com/apppasswords
#    get a 16 character "app password"
smtp <- server(host = "smtp.gmail.com",
               port = 587,
               username = "you@gmail.com",
               password = "REPLACE WITH 16 CHARACTER APP PASSWORD")

# Verizon:  5551234567@vtext.com
# AT&T:     5551234567@txt.att.net
# T-Mobile: 5551234567@tmomail.net
email <- envelope() |>
   from("you@gmail.com") |>
   to("5551234567@tmomail.net") |>
   text("Come back! Your movie data is ready!")

smtp(email, verbose = TRUE)
```

Note that the password here is in plain text so do not try this on a public computer. R also saves your history so even if it is not on the screen it might be saved somewhere else on the computer.

4 Parallel Computing

Since 1965 Moore's Law has predicted the power of computation over time. Moore's Law predicted the doubling of transistors about every two years. Moore's prediction has held true for decades. However, to get that speed the transistors were made smaller and smaller. Moore's Law cannot continue indefinitely. The diameter of a silicon atom is 0.2nm. Transistors today contain less than 70 atoms and some transistor dimensions are between 10nm and 40nm. Since 2012, computing power has not changed greatly signaling that we might be getting close to the end of Moore's Law, at least with silicon-based computing. What has changed is the widespread use of multicore processors. Rather than having a single processor, a typical laptop might have an 8 or 16 core processor (meaning they have 8 or 16 processors that share some resources like high speed memory).

R can guess how many cores your computer has on hand.

```
library(future)
library(doFuture)
parallelly::availableCores()
```

```
system
    16
```

Having access to multiple cores allows you to write scripts that send different tasks to different processors to work on simultaneously. While one processor is busy scraping the data for January 1st, the second can get to work on January 2nd, and another can work on January 3rd. All the processors will be fighting over the one connection you have to the internet, but they can `grep()` and `gsub()` at the same time other processors are working on other dates.

To write a script to work in parallel, you will need the `foreach` and `future` packages. Let's first test whether parallelization actually speeds things up. There are two `foreach` loops below. In both of them, each iteration of the loop does not really do anything except pause for 2 seconds. The first loop, which does not use parallelization, includes 10 iterations and so should take 20 seconds to run. The second `foreach` loop looks the same, except right before the `foreach` loop we have told R to make use of two of the computer's processors rather than the default of one processor. This should cause one processor to sleep for 2 seconds 10 times and the other processor to sleep for 2 seconds 10 times. In total this should take about 10 seconds.

```
library(foreach)

# should take 10*2=20 seconds
system.time( # time how long this takes
```

```r
  foreach(i=1:10) %do% # not in parallel
  {
     Sys.sleep(2)  # wait for 2 seconds
     return(i)
  }
)
```

```
   user  system elapsed 
   0.02    0.00   20.26 
```

```r
# set up R to use 2 cores
plan(multisession, workers = 2)
# tells %dopar% to use the plan's 2 cores
registerDoFuture()

# with two processors should take about 10 seconds
system.time(
  foreach(i=1:10) %dopar% # run in parallel
  {
    Sys.sleep(2)
    return(i)
  }
)
```

```
   user  system elapsed 
   0.15    0.07   10.36 
```

Sure enough, the parallel implementation was able to complete 20 seconds worth of sleeping in about 10 seconds. To set up code to run in parallel, the key steps are to set up the cores using `plan()` and to tell parallel `foreach()` to use that cluster of processors with `registerDoFuture()`. Note that the key difference between the two `foreach()` statements is that the first `foreach()` is followed by a `%do%` while the second is followed by a `%dopar%`. When `foreach()` sees the `%dopar%` it will check what was set up in the `registerDoFuture()` call and spread the computation among those cores.

Note that the `foreach()` differs a little bit in its syntax compared with our previous use of for-loops. While for-loops have the syntax `for(i in 1:10)` the syntax for `foreach()` looks like `foreach(i=1:10)` and is followed by a `%do%` or a `%dopar%`. Lastly, note that the final step inside the `{ }` following a `foreach()` is a `return()` statement. `foreach()` will take the returned values of each of the iterations and assemble them into a single list by default. In the following `foreach()` we have added `.combine=bind_rows` to the `foreach()` so that the final results will be stacked into one data frame, avoiding the need for a separate `bind_rows()` like we used previously.

Parallelization introduces some complications. If anything goes wrong in a parallelized script, then the whole `foreach()` fails. For example, let's say that after scraping movie data from 2000-2016 you briefly lose your internet connection. If this happens, then `scan()` fails and the whole `foreach()` will end with an error, tossing all of your completed work. To avoid this you need to either be sure you have a solid internet connection, or wrap the call to `scan()` in a `try()` and a `repeat` loop that is smart enough to wait a few seconds and try the scan again rather than fail completely.

This causes an error since this website does not exist (or not yet!).

```
res <- scan("https://www.jaywalkingIsNotACrime.org", what="", sep="\n")
```

```
Warning in file(file, "r"): URL 'https://www.jaywalkingIsNotACrime.org/':
status was 'Could not resolve hostname'
```

```
Error in `file()`:
! cannot open the connection to 'https://www.jaywalkingIsNotACrime.org'
```

```
# res does not exist
res
```

```
Error:
! object 'res' not found
```

If we wrap `scan()` with `try()`, then we can catch the error and write R code to gracefully handle the problem.

```
res <- try(scan("https://www.jaywalkingIsNotACrime.org", what="", sep="\n"),
           silent = TRUE) |>
   suppressWarnings()
is(res)
```

```
[1] "try-error"
```

```r
if(inherits(res, "try-error"))
{
   message("Could not find that website")
} else
{
   message("Found that website")
}
```

```
Could not find that website
```

With all this in mind, let's web scrape the movie data using multiple cores with `try()`/`repeat{}`. Typically, any attempts to print from inside a parallel `foreach()` do not appear in the console, since that print is running in a separate, parallel R session. The `progressr` package offers a way to print a progress bar to the console that also offers an estimated time to completion.

```r
# setup a Command-Line Interface progress bar
library(progressr)
handlers("cli")

plan(multisession, workers = 8)
registerDoFuture()

timeStart <- Sys.time() # record the starting time
# wrap the foreach inside the progress monitor
movieData <- with_progress(
{
   # create a progress bar how many total steps in the foreach loop
   p <- progressor(steps = length(dates2scrape))

   result <- foreach(iDate=1:length(dates2scrape),
                     .combine = bind_rows) %dopar%
   {
      # update progress bar
      p(paste("Working on", dates2scrape[iDate]))
      urlText <- paste0("https://www.the-numbers.com/box-office-chart/daily/",
                        gsub("-", "/", dates2scrape[iDate]))

      # retry up to 10 times with 10 sec sleep if fail
      tries <- 0
      repeat
```

```r
      {
         tries <- tries + 1
         a <- try(scan(urlText, what = "", sep = "\n",
                       fileEncoding = "UTF-8", quiet = TRUE),
                  silent = TRUE)
         if(!inherits(a, "try-error") || tries >= 10) break
         Sys.sleep(10)
      }
      # skip this date on persistent failure
      if(inherits(a, "try-error")) return(NULL)

      i <- grep('href="/movie/', a)
      i <- i[i < grep("chart-mobile",a)[1]]
      data0 <- data.frame(movie = gsub("<[^>]*>", "", a[i]),
                          gross = as.numeric(gsub("<[^>]*>|[$,]","",a[i+1])),
                          date  = dates2scrape[iDate])

      # replace empty movie names
      j <- which(data0$movie=="")
      if(length(j) > 0)
      {
         data0$movie[j] <- gsub('.*/movie/([^"]*)".*', "\\1", a[i[j]]) |>
            gsub("-", " ", x=_)
      }

      return(data0)
   }

   # the last object in with_progress() will be returned
   result
})

# calculate how long it took
timeEnd <- Sys.time()
timeEnd-timeStart
```

This code made use of 8 processors. Unlike our 2 second sleep example, this script may not be exactly 8 times faster. Each processor still needs to wait its turn in order to pull down its webpage from the internet. However, you should observe the parallel version finishing much sooner than the first version. In just a few lines of code and about 10 minutes of waiting, you now have 15 years worth of movie data.

Before moving on, let's do a final check that everything looks okay.

```r
nrow(movieData)
```

```
[1] 235874
```

```r
range(movieData$date)
```

```
[1] "2010-01-01" "2026-06-21"
```

```r
head(movieData)
```

```
                                   movie    gross       date
1                                 Avatar 25274008 2010-01-01
2                        Sherlock Holmes 14889882 2010-01-01
3 Alvin and the Chipmunks: The Squeakquel 12998264 2010-01-01
4                       It’s Complicated  7127425 2010-01-01
5                         The Blind Side  4554779 2010-01-01
6                          Up in the Air  4112263 2010-01-01
```

```r
tail(movieData)
```

```
                             movie  gross       date
235869           Stop! That! Train! 133797 2026-06-21
235870                     Pressure 120000 2026-06-21
235871 The Super Mario Galaxy Movie  35000 2026-06-21
235872              I Love Boosters  24871 2026-06-21
235873                 Unidentified   4000 2026-06-21
235874                   Deep Water    217 2026-06-21
```

Check for movie names with HTML codes.

```r
i <- grep("&[#0-9A-Za-z]+;", movieData$movie)
movieData$movie[i] |> unique()
```

```
  [1] "The Slammin&#039; Salmon"
  [2] "Le combat dans l&#039;île"
  [3] "Valentine&#039;s Day"
  [4] "Percy Jackson &amp; the Olympians: The Lightning Thief"
  [5] "She&#039;s Out of My League"
```

```
 [6] "George A. Romero&#039;s Survival of the Dead"
 [7] "Winter&#039;s Bone"
 [8] "Gangster&#039;s Paradise: Jerusalema"
 [9] "The Sorcerer&#039;s Apprentice"
[10] "Cats &amp; Dogs: The Revenge of Kitty Galore"
[11] "The People I&#039;ve Slept With"
[12] "Mao&#039;s Last Dancer"
[13] "L&#039;armée du crime"
[14] "It&#039;s Kind of a Funny Story"
[15] "Today&#039;s Special"
[16] "The Warrior&#039;s Way"
[17] "Saint Misbehavin&#039;: The Wavy Gravy Movie"
[18] "Barney&#039;s Version"
[19] "Now &amp; Later"
[20] "The Girl Who Kicked the Hornet&#039;s Nest"
[21] "Journal d&#039;un curé de campagne"
[22] "Madea&#039;s Big Happy Family"
[23] "I&#039;m Not Jesus Mommy"
[24] "Meek&#039;s Cutoff"
[25] "L&#039;Amour Fou"
[26] "Henry&#039;s Crime"
[27] "HEY BOO: Harper Lee and &quot;To Kill a Mockingbird&quot;"
[28] "Mr. Popper&#039;s Penguins"
[29] "Conan O&#039;Brien Can&#039;t Stop"
[30] "Sarah&#039;s Key"
[31] "The Devil&#039;s Double"
[32] "Don&#039;t Be Afraid of the Dark "
[33] "Beats, Rhymes &amp; Life: The Travels of a Tribe Called Quest"
[34] "I&#039;m Glad My Mother is Alive"
[35] "I Don&#039;t Know How She Does It"
[36] "What&#039;s Your Number?"
[37] "A Very Harold &amp; Kumar 3D Christmas"
[38] "Tyler Perry&#039;s Good Deeds"
[39] "Dr. Seuss&#039; The Lorax"
[40] "What to Expect When You&#039;re Expecting"
[41] "Madagascar 3: Europe&#039;s Most Wanted"
[42] "BRO&#039;"
[43] "Tyler Perry&#039;s Madea&#039;s Witness Protection"
[44] "Hit &amp; Run"
[45] "Won&#039;t Back Down"
[46] "&#039;Luv Shuv Tey Chicken Khurana"
[47] "Hansel &amp; Gretel: Witch Hunters"
[48] "Free Angela &amp; All Political Prisoners"
```

```
[49] "Pain &amp; Gain"
[50] "Dead Man&#039;s Burden"
[51] "Lee Daniels&#039; The Butler"
[52] "The World&#039;s End"
[53] "You&#039;re Next"
[54] "Jayne Mansfield&#039;s Car"
[55] "Romeo &amp; Juliet"
[56] "I&#039;m in Love with a Church Girl"
[57] "Return to Nuke &#039;Em High Volume 1"
[58] "Devil&#039;s Due"
[59] "Ernest &amp; Celestine"
[60] "Mr. Peabody &amp; Sherman"
[61] "Tyler Perry&#039;s The Single Moms Club"
[62] "Child&#039;s Pose"
[63] "Water &amp; Power"
[64] "Kirk Cameron&#039;s Saving Christmas"
[65] "The Devil&#039;s Violinist"
[66] "While We&#039;re Young"
[67] "I&#039;ll See You in My Dreams"
[68] "Love &amp; Mercy"
[69] "Jimmy&#039;s Hall"
[70] "Dragon Ball Z: Resurrection &quot;F&quot;"
[71] "Kahlil Gibran&#039;s The Prophet"
[72] "She&#039;s Funny That Way"
[73] "Hell &amp; Back"
[74] "This Isn&#039;t Funny"
[75] "Kapoor &amp; Sons"
[76] "Elvis &amp; Nixon"
[77] "Love &amp; Friendship"
[78] "My Love, Don&#039;t Cross That River"
[79] "Professor Marston &amp; The Wonder Women"
[80] "Gosnell: The Trial of America&#039;s Biggest Serial Killer"
[81] "The Big Bad Fox &amp; Other Tales"
[82] "Beauty &amp; the Beholder"
[83] "Holmes &amp; Watson"
[84] "Faith, Hope &amp; Love"
[85] "Marianne &amp; Leonard: Words of Love"
[86] "PreviewsFast &amp; Furious Presents: Hobbs &amp; Shaw"
[87] "Fast &amp; Furious Presents: Hobbs &amp; Shaw"
[88] "Vita &amp; Virginia"
[89] "Queen &amp; Slim"
[90] "PreviewsGretel &amp; Hansel"
[91] "Gretel &amp; Hansel"
```

```
 [92] "Bill &amp; Ted Face the Music"
 [93] "The Emperor&#039;s New Groove"
 [94] "Dating &amp; New York"
 [95] "Green Ghost &amp; The Masters Of The Stone"
 [96] "My Donkey, My Lover &amp; I"
 [97] "Mack &amp; Rita"
 [98] "Gigi &amp; Nate"
 [99] "The Chosen Season 3: Episodes 1 &amp; 2"
[100] "PreviewsDungeons &amp; Dragons: Honor Among Thieves"
[101] "Dungeons &amp; Dragons: Honor Among Thieves"
[102] "The Melt Goes on Forever: The Art &amp; Times of David Hammons"
[103] "Ernest &amp; Celestine: A Trip to Gibberitia"
[104] "Mother Teresa &amp; Me"
[105] "PreviewsThe Hunger Games: The Ballad of Songbirds &amp; Snakes"
[106] "The Hunger Games: The Ballad of Songbirds &amp; Snakes"
[107] "Deer Camp &#039;86"
[108] "Jatt &amp; Juliet 3"
[109] "PreviewsDeadpool &amp; Wolverine"
[110] "Deadpool &amp; Wolverine"
[111] "Sight &amp; Sound Presents: Daniel LIVE"
[112] "The Cowboy &amp; the Queen"
[113] "Howl&#039;s Moving Castle"
[114] "Love &amp; Pop"
[115] "Pride &amp; Prejudice"
[116] "Sod &amp; Stubble"
[117] "Juliet &amp; Romeo"
[118] "PreviewsLilo &amp; Stitch"
[119] "Lilo &amp; Stitch"
[120] "Tim Travers &amp; the Time Traveler’s Paradox"
[121] "given The Double Album: Hiiragi mix &amp; To the Sea Double Feature"
[122] "Hearts of Darkness: A Filmmaker&#039;s Apocalypse"
[123] "Sight &amp; Sound Presents: NOAH Live!"
[124] "Truth &amp; Treason"
[125] "PreviewsYou, Me &amp; Tuscany"
[126] "You, Me &amp; Tuscany"
```

Those HTML characters in movie titles are annoying to look at. Let’s fix it now.

```r
# change HTML codes to something prettier
movieData <- movieData |>
  mutate(movie = textutils::HTMLdecode(movie))
```

```r
# check that it worked
movieData$movie[i] |> unique()
```

```
 [1] "The Slammin' Salmon"
 [2] "Le combat dans l'île"
 [3] "Valentine's Day"
 [4] "Percy Jackson & the Olympians: The Lightning Thief"
 [5] "She's Out of My League"
 [6] "George A. Romero's Survival of the Dead"
 [7] "Winter's Bone"
 [8] "Gangster's Paradise: Jerusalema"
 [9] "The Sorcerer's Apprentice"
[10] "Cats & Dogs: The Revenge of Kitty Galore"
[11] "The People I've Slept With"
[12] "Mao's Last Dancer"
[13] "L'armée du crime"
[14] "It's Kind of a Funny Story"
[15] "Today's Special"
[16] "The Warrior's Way"
[17] "Saint Misbehavin': The Wavy Gravy Movie"
[18] "Barney's Version"
[19] "Now & Later"
[20] "The Girl Who Kicked the Hornet's Nest"
[21] "Journal d'un curé de campagne"
[22] "Madea's Big Happy Family"
[23] "I'm Not Jesus Mommy"
[24] "Meek's Cutoff"
[25] "L'Amour Fou"
[26] "Henry's Crime"
[27] "HEY BOO: Harper Lee and \"To Kill a Mockingbird\""
[28] "Mr. Popper's Penguins"
[29] "Conan O'Brien Can't Stop"
[30] "Sarah's Key"
[31] "The Devil's Double"
[32] "Don't Be Afraid of the Dark "
[33] "Beats, Rhymes & Life: The Travels of a Tribe Called Quest"
[34] "I'm Glad My Mother is Alive"
[35] "I Don't Know How She Does It"
[36] "What's Your Number?"
[37] "A Very Harold & Kumar 3D Christmas"
[38] "Tyler Perry's Good Deeds"
[39] "Dr. Seuss' The Lorax"
```

```
[40] "What to Expect When You're Expecting"
[41] "Madagascar 3: Europe's Most Wanted"
[42] "BRO'"
[43] "Tyler Perry's Madea's Witness Protection"
[44] "Hit & Run"
[45] "Won't Back Down"
[46] "'Luv Shuv Tey Chicken Khurana"
[47] "Hansel & Gretel: Witch Hunters"
[48] "Free Angela & All Political Prisoners"
[49] "Pain & Gain"
[50] "Dead Man's Burden"
[51] "Lee Daniels' The Butler"
[52] "The World's End"
[53] "You're Next"
[54] "Jayne Mansfield's Car"
[55] "Romeo & Juliet"
[56] "I'm in Love with a Church Girl"
[57] "Return to Nuke 'Em High Volume 1"
[58] "Devil's Due"
[59] "Ernest & Celestine"
[60] "Mr. Peabody & Sherman"
[61] "Tyler Perry's The Single Moms Club"
[62] "Child's Pose"
[63] "Water & Power"
[64] "Kirk Cameron's Saving Christmas"
[65] "The Devil's Violinist"
[66] "While We're Young"
[67] "I'll See You in My Dreams"
[68] "Love & Mercy"
[69] "Jimmy's Hall"
[70] "Dragon Ball Z: Resurrection \"F\""
[71] "Kahlil Gibran's The Prophet"
[72] "She's Funny That Way"
[73] "Hell & Back"
[74] "This Isn't Funny"
[75] "Kapoor & Sons"
[76] "Elvis & Nixon"
[77] "Love & Friendship"
[78] "My Love, Don't Cross That River"
[79] "Professor Marston & The Wonder Women"
[80] "Gosnell: The Trial of America's Biggest Serial Killer"
[81] "The Big Bad Fox & Other Tales"
[82] "Beauty & the Beholder"
```

```
 [83] "Holmes & Watson"
 [84] "Faith, Hope & Love"
 [85] "Marianne & Leonard: Words of Love"
 [86] "PreviewsFast & Furious Presents: Hobbs & Shaw"
 [87] "Fast & Furious Presents: Hobbs & Shaw"
 [88] "Vita & Virginia"
 [89] "Queen & Slim"
 [90] "PreviewsGretel & Hansel"
 [91] "Gretel & Hansel"
 [92] "Bill & Ted Face the Music"
 [93] "The Emperor's New Groove"
 [94] "Dating & New York"
 [95] "Green Ghost & The Masters Of The Stone"
 [96] "My Donkey, My Lover & I"
 [97] "Mack & Rita"
 [98] "Gigi & Nate"
 [99] "The Chosen Season 3: Episodes 1 & 2"
[100] "PreviewsDungeons & Dragons: Honor Among Thieves"
[101] "Dungeons & Dragons: Honor Among Thieves"
[102] "The Melt Goes on Forever: The Art & Times of David Hammons"
[103] "Ernest & Celestine: A Trip to Gibberitia"
[104] "Mother Teresa & Me"
[105] "PreviewsThe Hunger Games: The Ballad of Songbirds & Snakes"
[106] "The Hunger Games: The Ballad of Songbirds & Snakes"
[107] "Deer Camp '86"
[108] "Jatt & Juliet 3"
[109] "PreviewsDeadpool & Wolverine"
[110] "Deadpool & Wolverine"
[111] "Sight & Sound Presents: Daniel LIVE"
[112] "The Cowboy & the Queen"
[113] "Howl's Moving Castle"
[114] "Love & Pop"
[115] "Pride & Prejudice"
[116] "Sod & Stubble"
[117] "Juliet & Romeo"
[118] "PreviewsLilo & Stitch"
[119] "Lilo & Stitch"
[120] "Tim Travers & the Time Traveler’s Paradox"
[121] "given The Double Album: Hiiragi mix & To the Sea Double Feature"
[122] "Hearts of Darkness: A Filmmaker's Apocalypse"
[123] "Sight & Sound Presents: NOAH Live!"
[124] "Truth & Treason"
[125] "PreviewsYou, Me & Tuscany"
```

```
[126] "You, Me & Tuscany"
```

It is probably wise at this point to save `movieData` so that you will not have to rerun this if you mess up your dataset. With `movieData` saved you should feel free to test out your ideas. You can always `load("movieData.RData")` if you make a mistake.

```
save(movieData, file="movieData.RData", compress=TRUE)
```

5 Fun With Movie Data

You can use the dataset to answer questions such as "which movie yielded the largest gross?"

```
movieData |> slice_max(gross)
```

```
              movie     gross       date
1 Avengers: Endgame 157461641 2019-04-26
```

Which ten movies had the largest total gross during the period this dataset covers?

```
movieData |>
   summarize(gross=sum(gross), .by=movie) |>
   slice_max(gross, n=10)
```

```
                                    movie     gross
1  Star Wars Ep. VII: The Force Awakens 935642689
2                     Avengers: Endgame 858373000
3                 Spider-Man: No Way Home 804793477
4                     Top Gun: Maverick 718732821
5                         Black Panther 700059566
6                Avatar: The Way of Water 688459501
7                  Avengers: Infinity War 678815482
8                          Inside Out 2 652980194
9                        Jurassic World 652198010
10                        The Lion King 637188286
```

Which days of the week yielded the largest total gross?

```
movieData |>
   mutate(weekday=wday(date, label=TRUE)) |>
   summarize(gross=sum(gross), .by=weekday) |>
   arrange(desc(gross))
```

```
  weekday       gross
1     Sat 41168630293
2     Fri 35201187015
3     Sun 28872202888
4     Thu 13336097315
5     Tue 12901470627
6     Mon 12187074394
7     Wed 10755702057
```

5.1 Adjusting for Inflation

As you may have noticed, the price of a movie ticket keeps increasing (along with everything else!). The Bureau of Labor Statistics (BLS) tracks the Consumer Price Index (CPI) for "Admission to movies, theaters, and concerts" (series `CUUR0000SS62031`). The CPI is an index, not a dollar price — the base period (1998) equals 100, so a value of 240 means tickets cost 2.4× what they did in the late 1990s. What matters for us is the *ratio* between years, which gives us the inflation adjustment factor. That factor will vary by year. It will tell you how much you need to multiply, say, a ticket purchased in 2010 so that it equates to 2026 prices.

The BLS public API limits unauthenticated requests to 10 years of data per call, so we make two requests and combine them. The API returns monthly values (periods `M01`–`M12`). We will average those to get one index value per year.

```
library(httr2)

# max 10 years per call (if run without API key)
a <- request("https://api.bls.gov/publicAPI/v2/timeseries/data/") |>
   req_body_json(list(seriesid  = list("CUUR0000SS62031"),
                      startyear = 2010,
                      endyear   = 2019)) |>
   req_perform() |>
   resp_body_json() |>
   _$Results$series[[1]]$data

b <- request("https://api.bls.gov/publicAPI/v2/timeseries/data/") |>
```

```
   req_body_json(list(seriesid  = list("CUUR0000SS62031"),
                      startyear = 2020,
                      endyear   = 2026)) |>
   req_perform() |>
   resp_body_json() |>
   _$Results$series[[1]]$data

inflation <- c(a,b)  |>
   lapply(\(x) data.frame(year   = as.integer(x$year),
                          period = x$period,
                          value  = as.numeric(x$value))) |>
   bind_rows() |>
   # one month in 2025 missing due to gov't shutdown
   summarize(cpiMovie = mean(value, na.rm=TRUE), .by = year) |>
   arrange(desc(year)) |>
   mutate(adjustment = cpiMovie[1] / cpiMovie)
```

```
Warning in data.frame(year = as.integer(x$year), period = x$period, value =
as.numeric(x$value)): NAs introduced by coercion
```

Now we link each movie to our inflation factor table to compute ticket sales adjusted to 2026 prices.

```
movieData <- movieData |>
   mutate(year=year(date)) |>
   left_join(inflation, join_by(year==year)) |>
   mutate(grossAdj=gross*adjustment) |>
   select(-cpiMovie, -adjustment)

movieData |>
  summarize(grossAdj=sum(grossAdj), .by=movie) |>
  slice_max(grossAdj, n=10)
```

```
                                movie   grossAdj
1  Star Wars Ep. VII: The Force Awakens 1342790345
2                    Avengers: Endgame 1127403684
3              Spider-Man: No Way Home  994098727
4                       Jurassic World  946032392
5                         The Avengers  944786642
6                        Black Panther  936184030
7               Avengers: Infinity War  907774487
```

```
8                          The Lion King  858594577
9                      Top Gun: Maverick  851487183
10    Star Wars Ep. VIII: The Last Jedi  843569212
```

The Avengers and *Star Wars: The Last Jedi* join the top 10 list and *Avatar* and *Inside Out 2* are out.

Now that you have movie data and, in a previous section, you assembled Chicago crime data, combine the two datasets so that you can answer the question "What happens to crime when big movies come out?"